METRIC
MIL-DTL-32598
5 March 2018
SUPERSEDING
O-S-639B
27 August 1984
(See 6.7)

DETAIL SPECIFICATION

SODIUM PHOSPHATE, DIBASIC, ANHYDROUS, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers technical grade anhydrous dibasic sodium phosphate primarily intended for boiler water treatment (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of this document are available online at http://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D460 - Standard Test Methods for Sampling and Chemical Analysis of Soaps and Soap Products

ASTM D820 - Standard Test Methods for Chemical Analysis of Soaps Containing Synthetic Detergents

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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ASTM D1193 - Standard Specification for Reagent Water

ASTM D1293 - Standard Test Methods for pH of Water

(Copies of these documents are available online at www.astm.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1331 - Zero Acceptance Number Sampling Plans

(Copies of this document are available online at www.asq.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Anhydrous dibasic sodium phosphate (Na_2HPO_4) shall be in the form of a white powder, flakes, or granular.

3.2 pH value. The pH of a 1-percent solution shall be between 8.7 and 9.3 (see 4.2.1).

3.3 Composition. The composition shall be in accordance with table I.

TABLE I. Composition.

Characteristic	Maximum percent	Minimum percent	Test
Na_2HPO_4	----	97.5	4.2.2.1
Water	2.0	----	4.2.2.2
Chlorides as sodium chloride	0.2	----	4.2.2.3
Insoluble matter	0.15	----	4.2.2.4
NOTE: "----" indicates that the test does not apply.			

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall be performed in accordance with inspection provisions set forth herein. The characteristics set forth in table I, when tested in accordance with 4.2, shall constitute minimum inspections to be performed. As specified (see 6.2), sampling shall be in accordance with MIL-STD-1916, ASQ H1331, or a comparable zero-based sampling plan. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material submitted for inspection and delivery at one time. Failure of any test, by any sample, shall be cause for rejection of the lot represented. Results shall be as specified (see 6.2).

4.2 Test methods and procedures.

4.2.1 pH value. A 1-percent solution of the sample shall be prepared using reagent water conforming to ASTM D1193, type II. The pH of this solution shall be measured in accordance with ASTM D1293.

4.2.2 Composition.

4.2.2.1 Na_2HPO_4. The test shall be conducted as specified in 4.2.2.1.1 except that in case of dispute, the referee test specified in 4.2.2.1.2 shall be used.

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4.2.2.1.1 Composition test. Transfer a 2.5±0.001-gram sample to a 400-milliliter (mL) beaker. Dissolve in 100 mL of reagent water conforming to ASTM D1193, type II. Add 15 mL concentrated hydrochloric acid (specific gravity 1.19). Boil the covered solution for 30 minutes. Cool to room temperature, and dilute to 200 mL. Adjust to 4.3 pH in accordance with ASTM D1293, using 50-percent sodium hydroxide (NaOH) solution (approximately 50 grams per 100-mL solution) until reaching about 3.5 pH, and making the final adjustment with standard NaOH solution (0.5 normal [N]). Then titrate to 8.8 pH with the standard 0.5 N NaOH solution, recording the titration between 4.3 and 8.8 pH. Standardize the electrometer against a pH 7 buffer solution immediately before use.

Calculation:

$$\text{Percent } Na_2HPO_4 = \frac{M \times N \times 14.196}{W}$$

Where:

W = weight of sample.

M = mL of NaOH solution used between 4.3 and 8.8 pH.

N = normality of NaOH solution.

4.2.2.1.2 Referee test. Weigh, to the nearest 0.1 milligram (mg), approximately 0.1 gram of the sample. Dissolve in reagent water conforming to ASTM D1193, type II and make up to 250 mL in a volumetric flask. Take a 50-mL aliquot of the solution and determine "Phosphates" in accordance with ASTM D820.

4.2.2.2 Water. The water content shall be determined in accordance with method B of ASTM D460.

4.2.2.3 Chlorides. A 0.2-gram sample shall be dissolved in 20 mL of reagent water conforming to ASTM D1193, type II and 3 mL of 7 N nitric acid and 1 mL of 0.1 N silver-nitrate solution added. Any turbidity shall be not greater than is produced by a solution of 0.002 percent sodium chloride of an equal volume with the same quantities of acid and silver-nitrate solution.

4.2.2.4 Insoluble matter. A 2.5-gram sample shall be dissolved in 500 mL of warm (60 to 65 °C) reagent water conforming to ASTM D1193, type II. If turbidity or undissolved matter is present, the sample shall be filtered through a weighed Gooch crucible, washing with five 10-mL portions of warm reagent water. The crucible and contents shall be dried to a constant weight by heating at 100 to 105 °C. Insoluble matter shall be weighed and the percentage calculated.

$$\text{Percent of insoluble} = \frac{\text{grams of residue}}{\text{grams of sample}} \times 100$$

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The anhydrous dibasic sodium phosphate covered in this specification is primarily intended for boiler water treatment and may also be used in water softening purposes and as an ingredient in compounded metal cleaners. The material is not intended for medicinal purposes. Water softened with this material is not intended for human consumption. Items acquired under this specification for military use are to be limited to the variety shown on the applicable military standard or military supply standard. Personnel of the military departments are requested to refer to these documents for guidance.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Requirement for the results obtained from all inspections performed (see 4.1).

c. Sampling plan in accordance with MIL-STD-1916 at verification level IV or ASQ H1331 at an Acceptance Quality Level of 1.0, or a comparable zero-based sampling plan (see 4.1).

d. Packaging requirements (see 5.1).

e. When Safety Data Sheets (SDS) are required (see 6.5).

6.3 Alternate names. Dibasic sodium phosphate is also known commercially as disodium phosphate, sodium hydrogen phosphate, and disodium orthophosphate.

6.4 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, DoD Shelf-Life Management Program. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.5 SDSs. When specified (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.6 National stock numbers (NSN). Table II provides a list of NSNs assigned which correspond to this specification. This list may not include all possible NSNs associated with this document.

TABLE II. NSNs.

NSN	Shelf-life code	Unit of issue
6810-00-264-6630	0 (does not expire)	100-lb bag
6810-00-584-4298	0 (does not expire)	25-lb drum

6.7 Use of existing stocks. This specification supersedes O-S-639. Current stocks of dibasic sodium phosphate made to O-S-639 may continue to be used for their respective purposes until stocks are exhausted. New acquisitions will be made in accordance with this specification.

6.8 Subject term (key word) listing.

Boiler chemicals

Disodium orthophosphate

Disodium phosphate

Metal cleaner

Sodium hydrogen phosphate

Water softener

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CONCLUDING MATERIAL

Custodians:
- Army – GL
- Navy – SH
- Air Force – 68
- DLA – GS

Review activities:
- Army – AV, EA, MI, MR, TE
- Navy – AS
- Air Force – 03, 11, 20
- DLA – GS8

Civil agency:
- GSA – FAS

Preparing activity:
- Navy – SH

(Project 6810-2017-011)

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